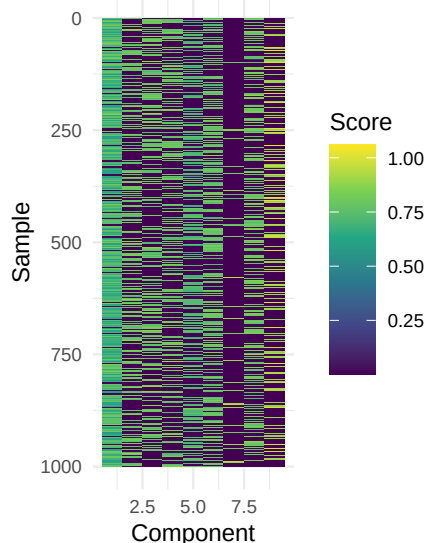


Denoising factor estimates: simulation 0

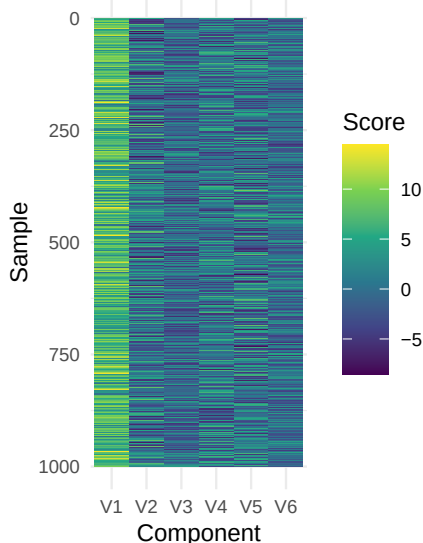
Simulation 0: GB + Laplace

Fitted factors = 9; RMSE = 0.1927



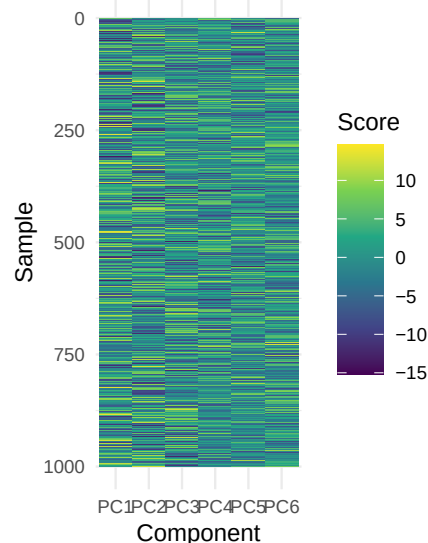
Simulation 0: cEBMF= cGB

Fitted factors = 4; RMSE = 0.1908



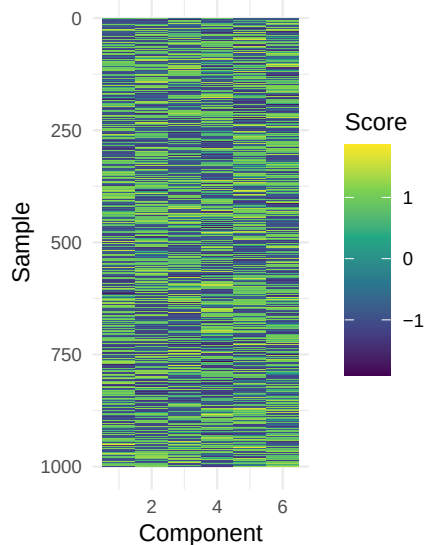
Simulation 0: oracle PCA

Oracle rank = 6; RMSE = 0.2432



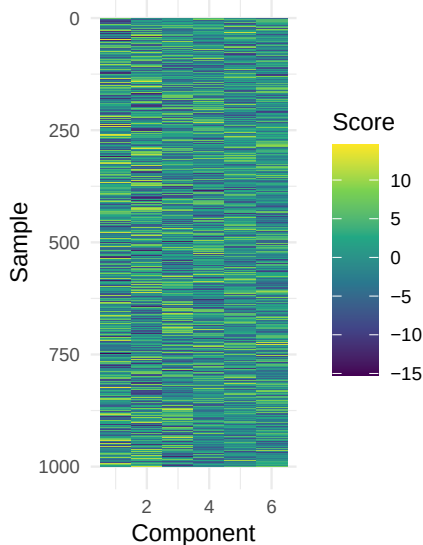
Simulation 0: oracle ICA

Oracle rank = 6; RMSE = 0.2432



Simulation 0: oracle-rank SoftImpute

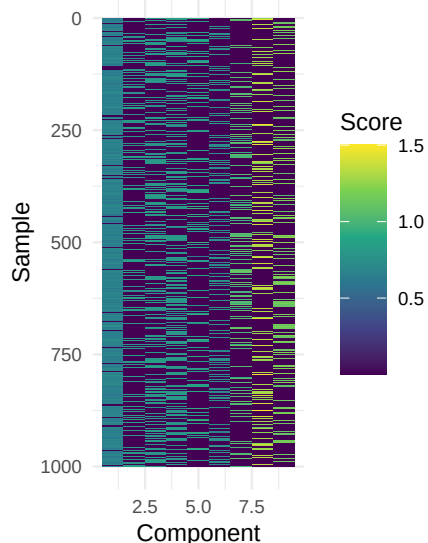
Oracle rank.max = 6; effective rank = 6; lambda = 1; RMSE = 0.2417



Denoising factor estimates: simulation 1

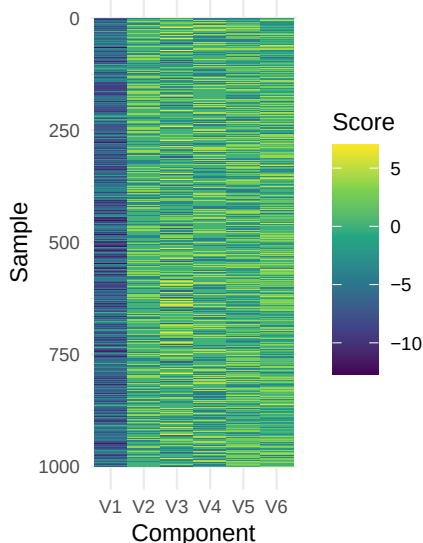
Simulation 1: GB + Laplace

Fitted factors = 9; RMSE = 0.1366



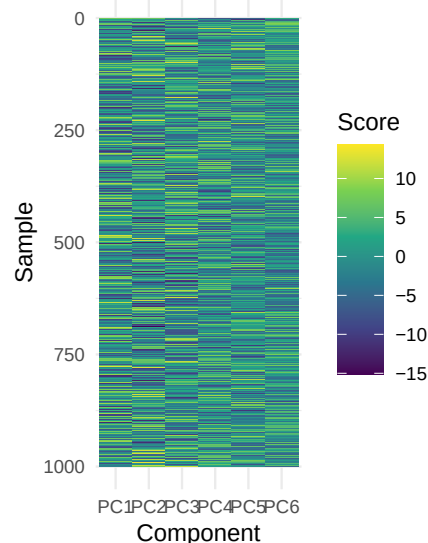
Simulation 1: cEBMF= cGB

Fitted factors = 4; RMSE = 0.1492



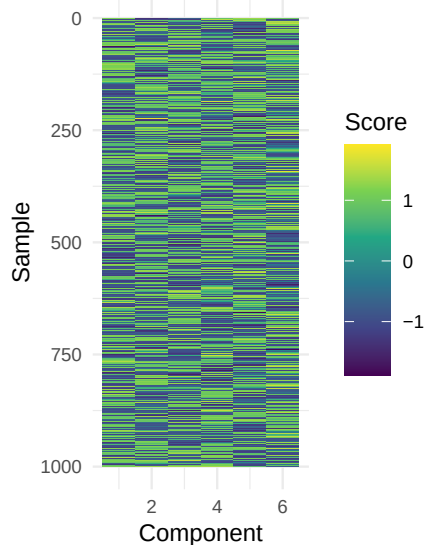
Simulation 1: oracle PCA

Oracle rank = 6; RMSE = 0.2449



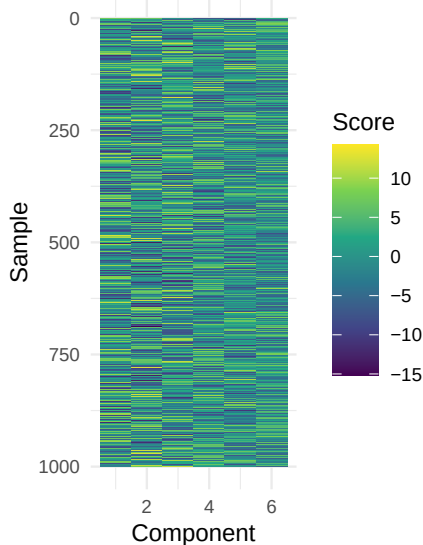
Simulation 1: oracle ICA

Oracle rank = 6; RMSE = 0.2449



Simulation 1: oracle-rank SoftImpute

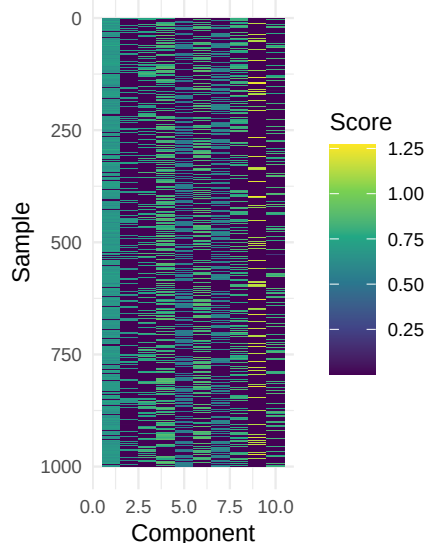
Oracle rank.max = 6; effective rank = 6; lambda = 1; RMSE = 0.2436



Denoising factor estimates: simulation 2

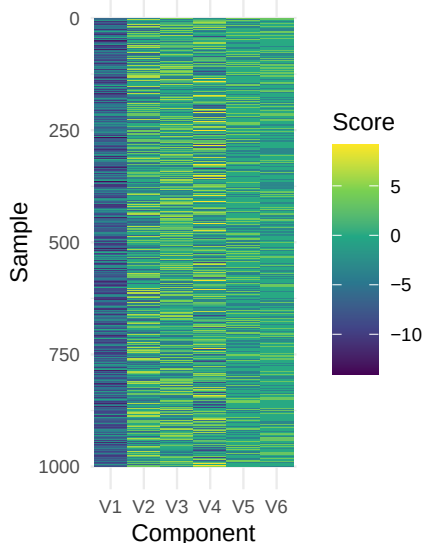
Simulation 2: GB + Laplace

Fitted factors = 10; RMSE = 0.151



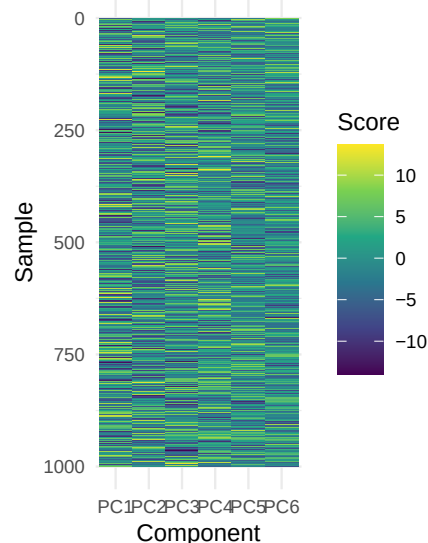
Simulation 2: cEBMF= cGB

Fitted factors = 4; RMSE = 0.1436



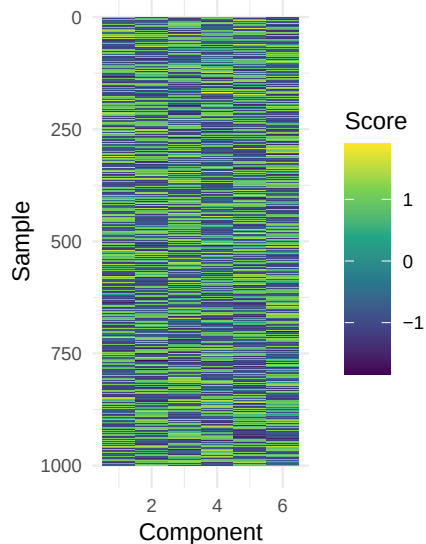
Simulation 2: oracle PCA

Oracle rank = 6; RMSE = 0.2421



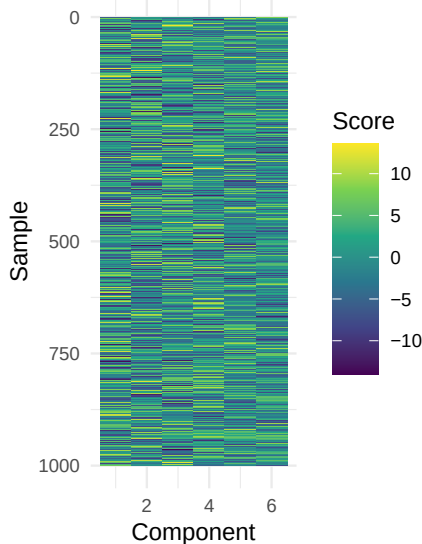
Simulation 2: oracle ICA

Oracle rank = 6; RMSE = 0.2421



Simulation 2: oracle-rank SoftImpute

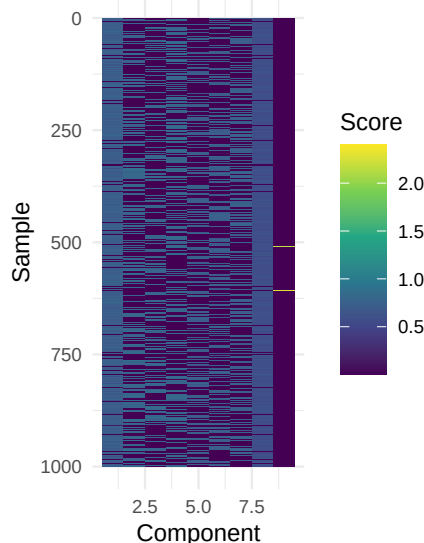
Oracle rank.max = 6; effective rank = 6; lambda = 1; RMSE = 0.2406



Denoising factor estimates: simulation 3

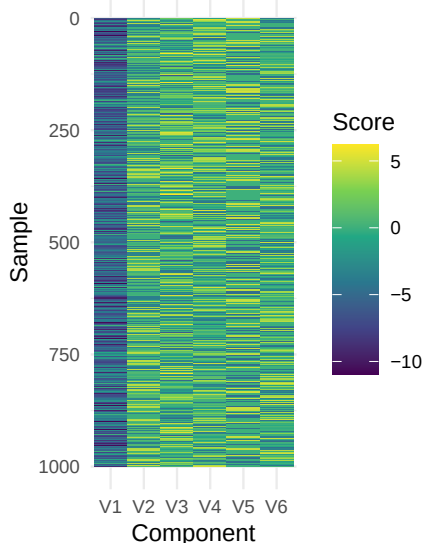
Simulation 3: GB + Laplace

Fitted factors = 9; RMSE = 0.1712



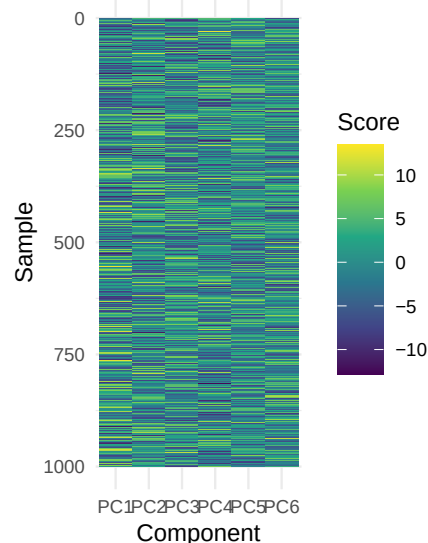
Simulation 3: cEBMF= cGB

Fitted factors = 4; RMSE = 0.1436



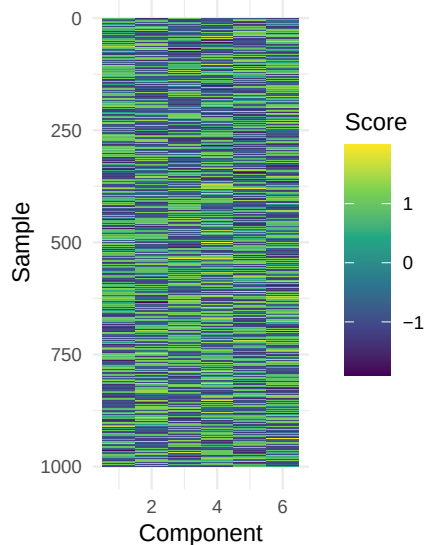
Simulation 3: oracle PCA

Oracle rank = 6; RMSE = 0.2425



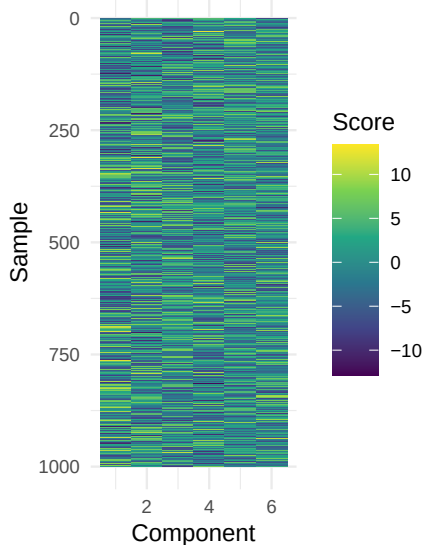
Simulation 3: oracle ICA

Oracle rank = 6; RMSE = 0.2425



Simulation 3: oracle-rank SoftImpute

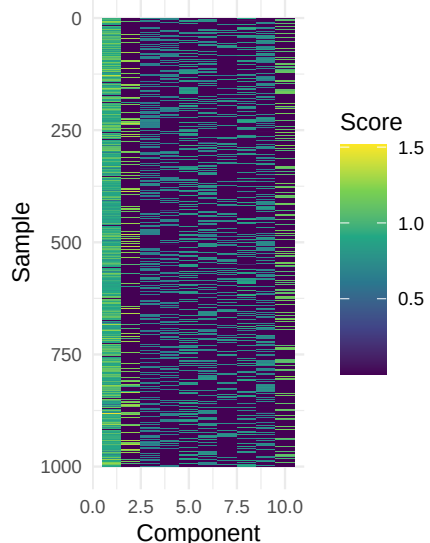
Oracle rank.max = 6; effective rank = 6; lambda = 1; RMSE = 0.2410



Denoising factor estimates: simulation 4

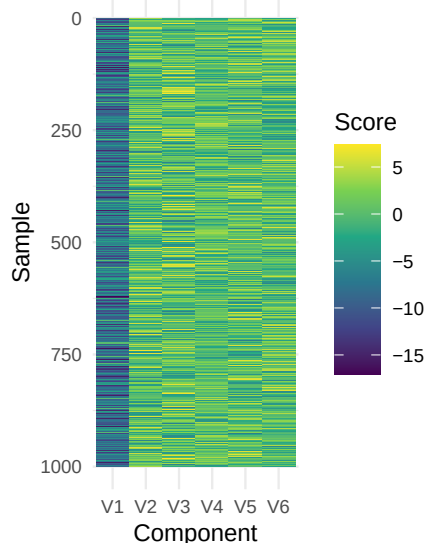
Simulation 4: GB + Laplace

Fitted factors = 10; RMSE = 0.235



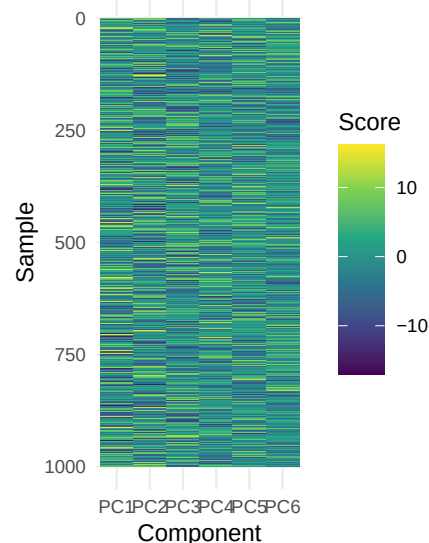
Simulation 4: cEBMF= cGB

Fitted factors = 4; RMSE = 0.1874



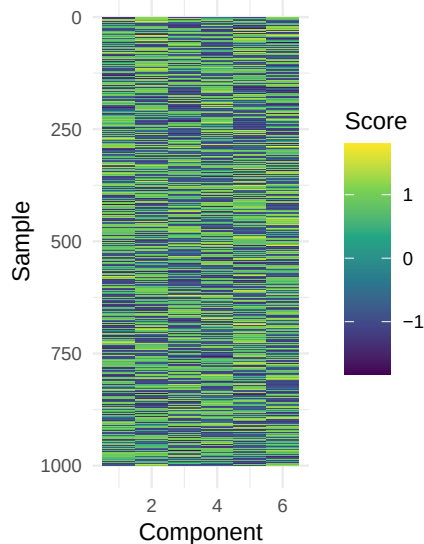
Simulation 4: oracle PCA

Oracle rank = 6; RMSE = 0.2468



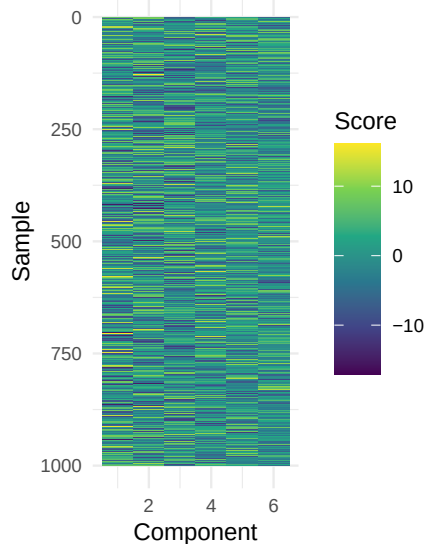
Simulation 4: oracle ICA

Oracle rank = 6; RMSE = 0.2468



Simulation 4: oracle-rank SoftImpute

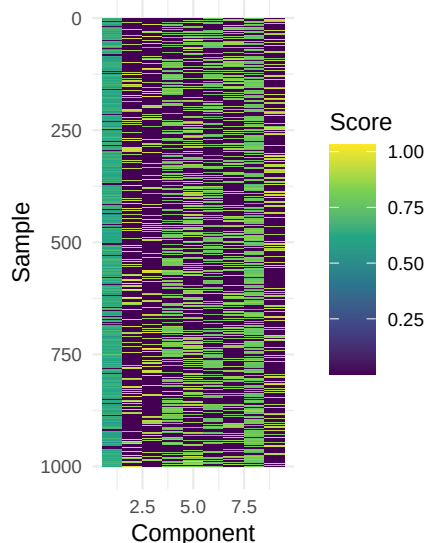
Oracle rank.max = 6; effective rank = 6; lambda = 1; RMSE = 0.2454



Denoising factor estimates: simulation 5

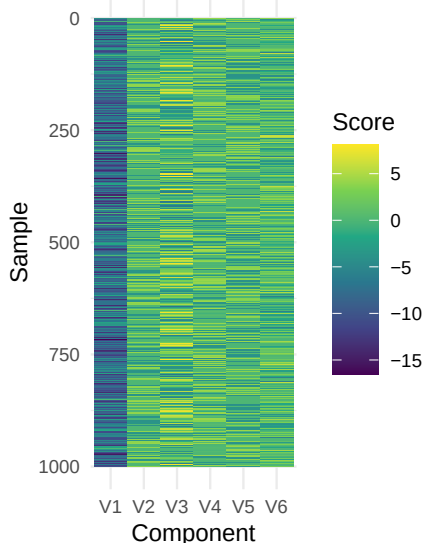
Simulation 5: GB + Laplace

Fitted factors = 9; RMSE = 0.1418



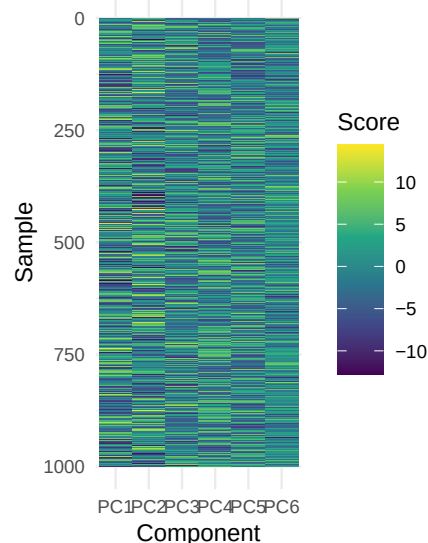
Simulation 5: cEBMF= cGB

Fitted factors = 4; RMSE = 0.1543



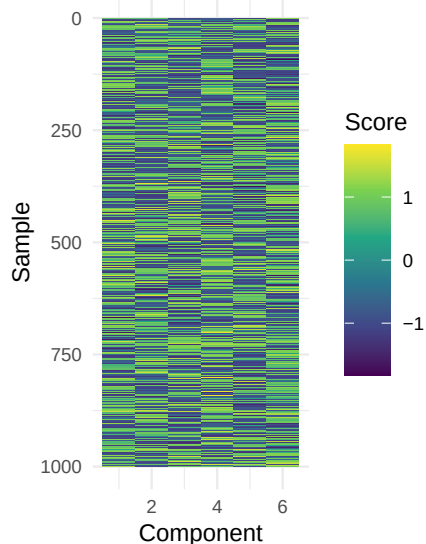
Simulation 5: oracle PCA

Oracle rank = 6; RMSE = 0.2401



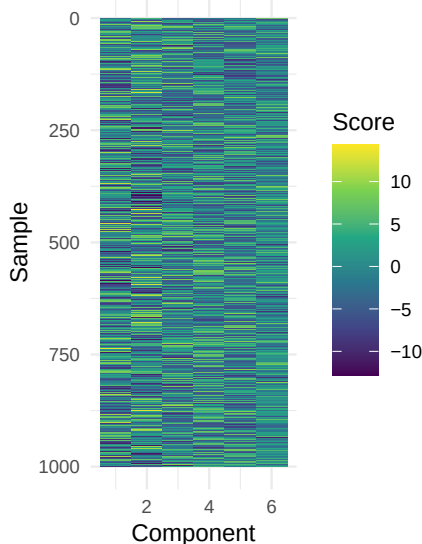
Simulation 5: oracle ICA

Oracle rank = 6; RMSE = 0.2401



Simulation 5: oracle-rank SoftImpute

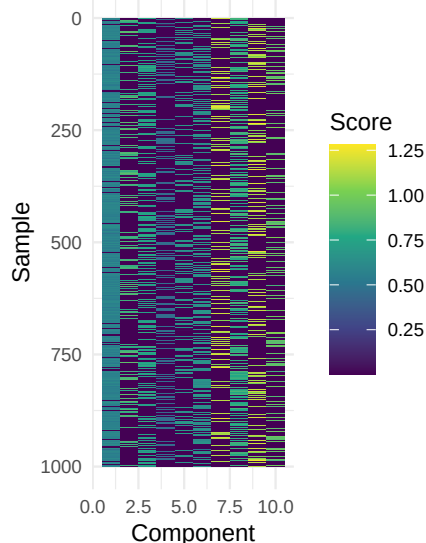
Oracle rank.max = 6; effective rank = 6; lambda = 1; RMSE = 0.2387



Denoising factor estimates: simulation 6

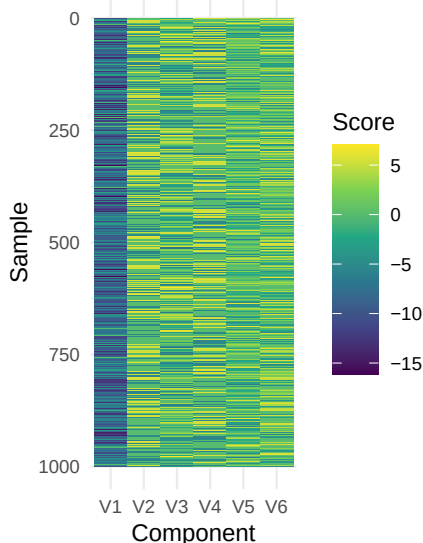
Simulation 6: GB + Laplace

Fitted factors = 10; RMSE = 0.141



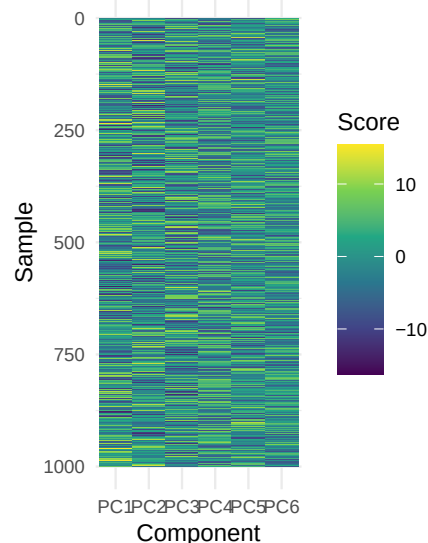
Simulation 6: cEBMF= cGB

Fitted factors = 4; RMSE = 0.166



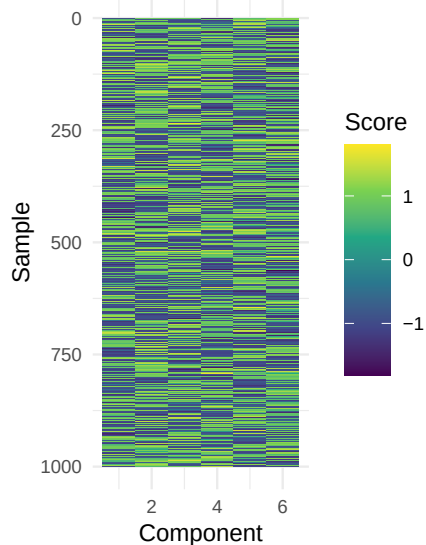
Simulation 6: oracle PCA

Oracle rank = 6; RMSE = 0.2408



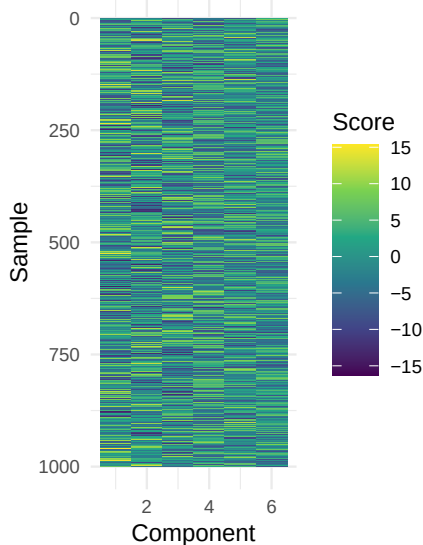
Simulation 6: oracle ICA

Oracle rank = 6; RMSE = 0.2408



Simulation 6: oracle-rank SoftImpute

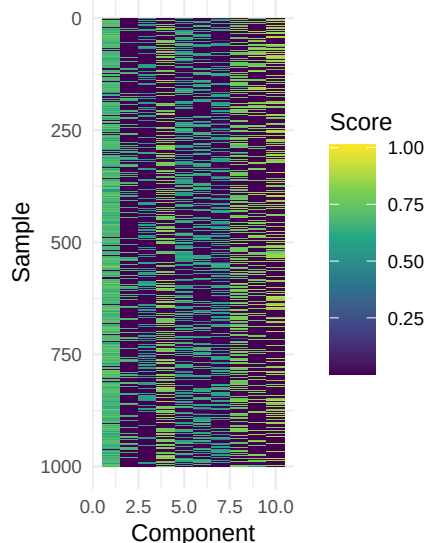
Oracle rank.max = 6; effective rank = 6; lambda = 1; RMSE = 0.2395



Denoising factor estimates: simulation 7

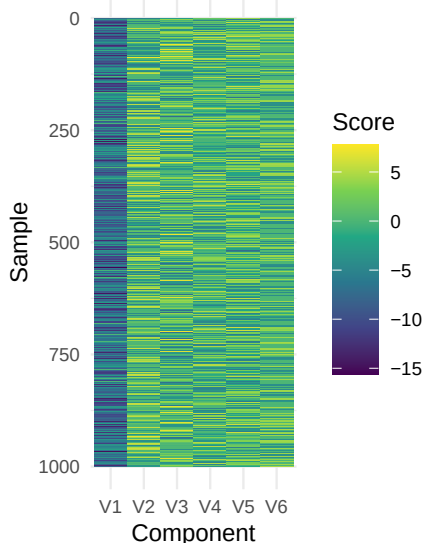
Simulation 7: GB + Laplace

Fitted factors = 10; RMSE = 0.225



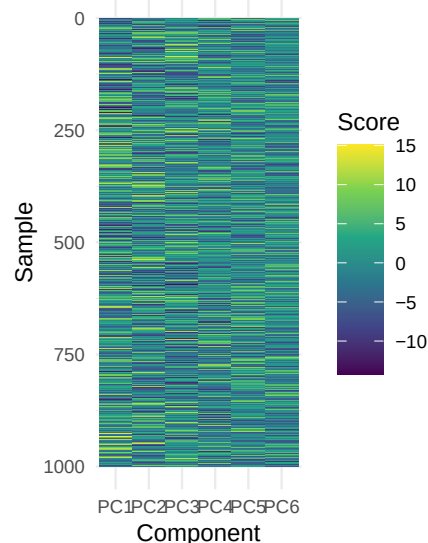
Simulation 7: cEBMF= cGB

Fitted factors = 4; RMSE = 0.1586



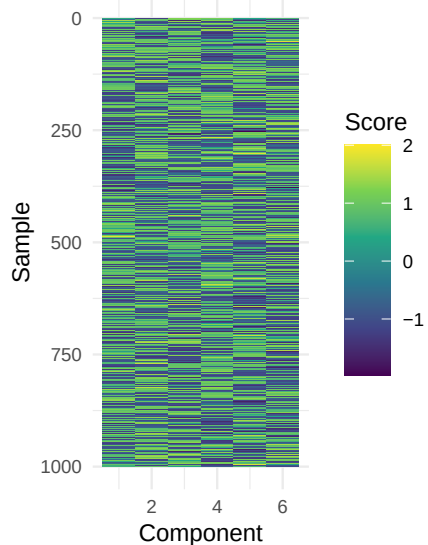
Simulation 7: oracle PCA

Oracle rank = 6; RMSE = 0.2477



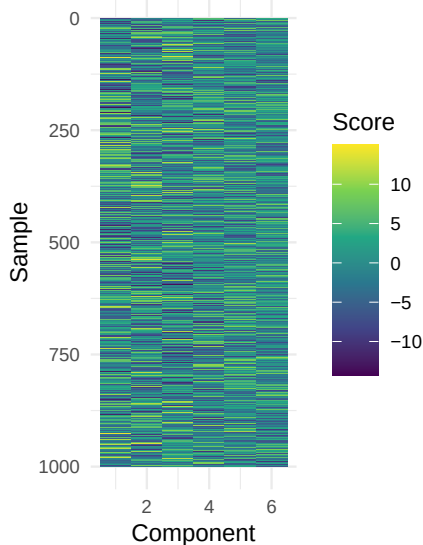
Simulation 7: oracle ICA

Oracle rank = 6; RMSE = 0.2477



Simulation 7: oracle-rank SoftImpute

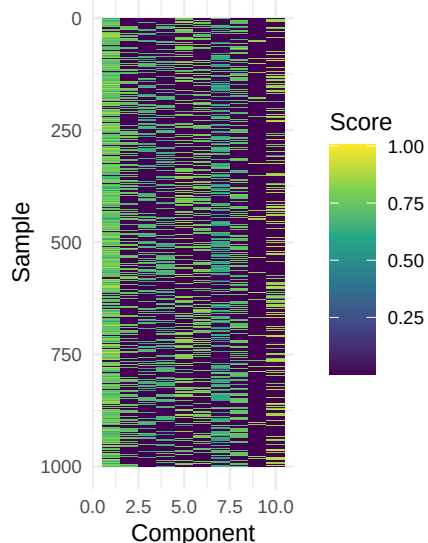
Oracle rank.max = 6; effective rank = 6; lambda = 1; RMSE = 0.2462



Denoising factor estimates: simulation 8

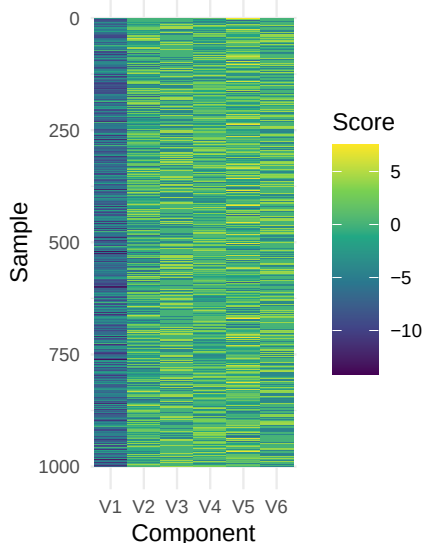
Simulation 8: GB + Laplace

Fitted factors = 10; RMSE = 0.203



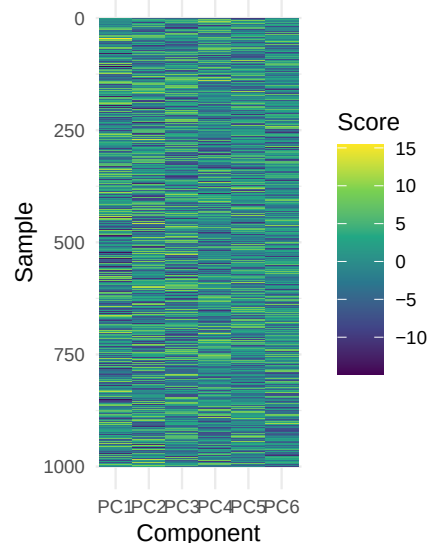
Simulation 8: cEBMF= cGB

Fitted factors = 4; RMSE = 0.1577



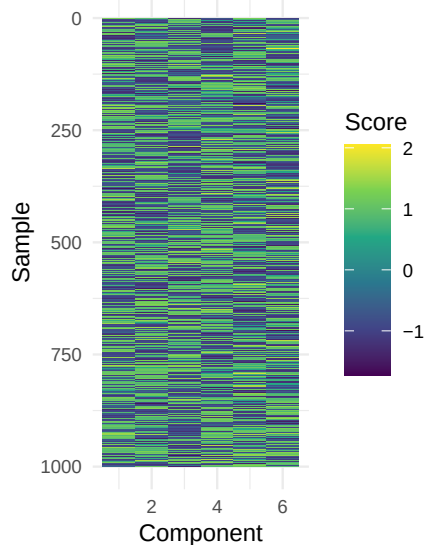
Simulation 8: oracle PCA

Oracle rank = 6; RMSE = 0.2434



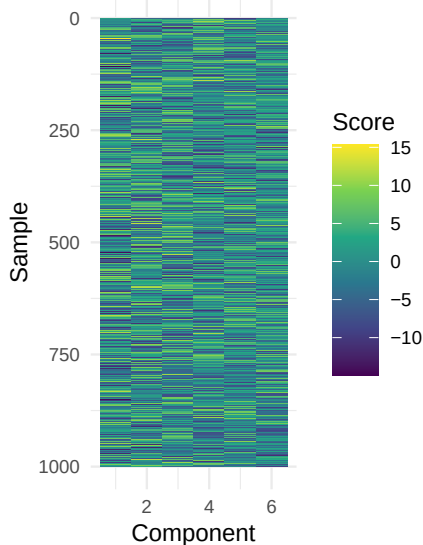
Simulation 8: oracle ICA

Oracle rank = 6; RMSE = 0.2434



Simulation 8: oracle-rank SoftImpute

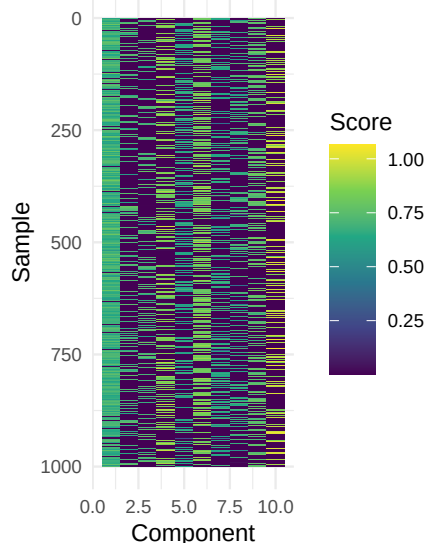
Oracle rank.max = 6; effective rank = 6; lambda = 1; RMSE = 0.2419



Denoising factor estimates: simulation 9

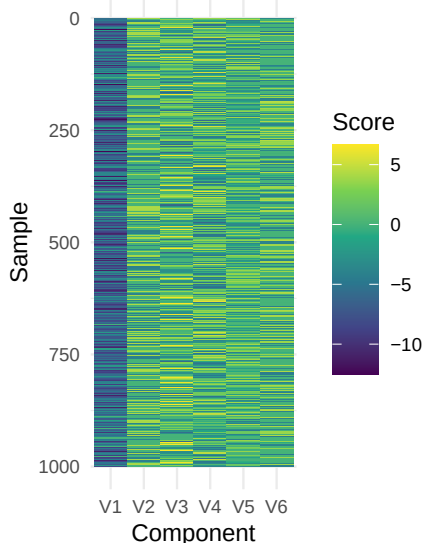
Simulation 9: GB + Laplace

Fitted factors = 10; RMSE = 0.182



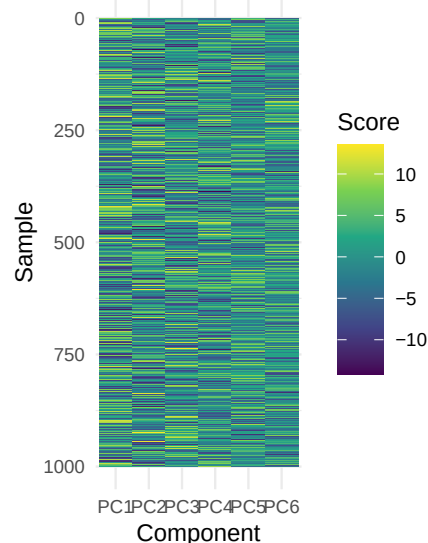
Simulation 9: cEBMF= cGB

Fitted factors = 4; RMSE = 0.1585



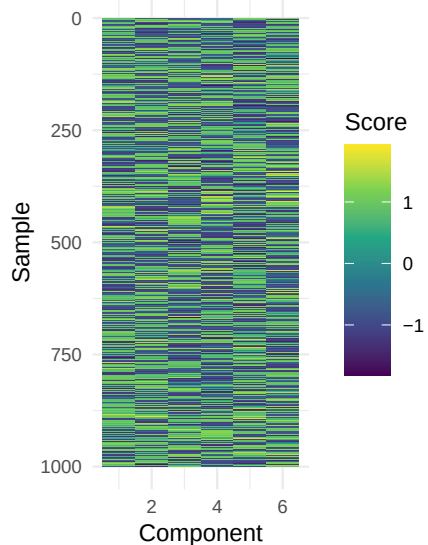
Simulation 9: oracle PCA

Oracle rank = 6; RMSE = 0.2438



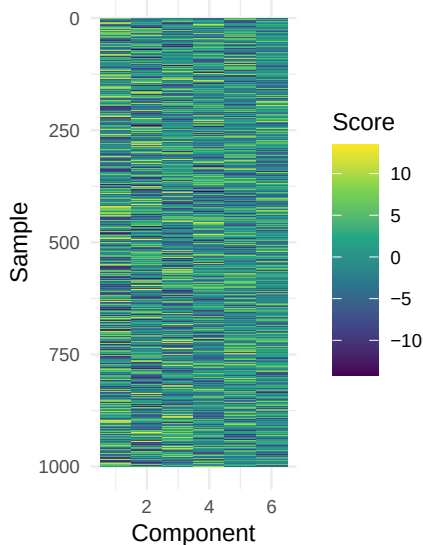
Simulation 9: oracle ICA

Oracle rank = 6; RMSE = 0.2438



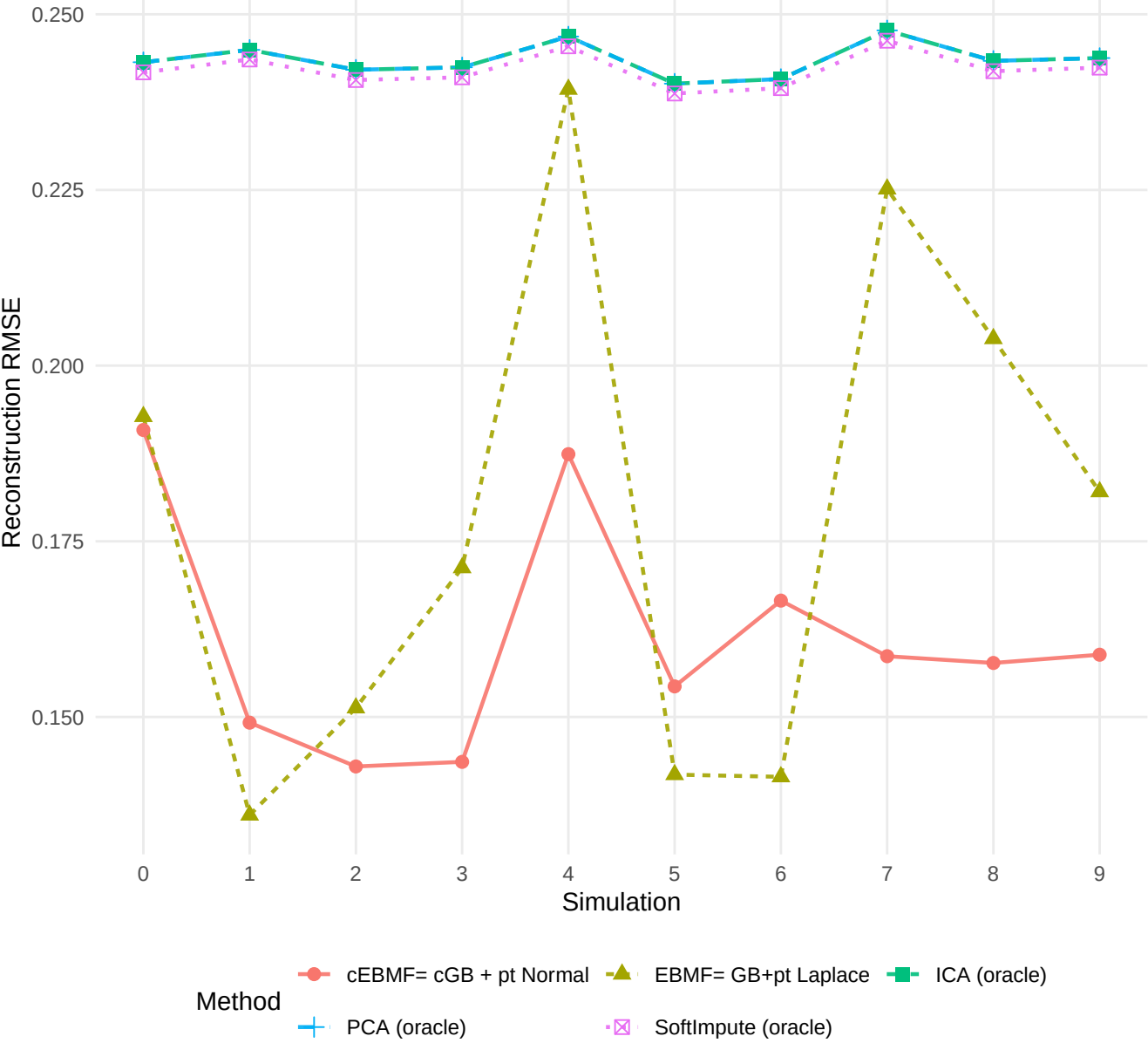
Simulation 9: oracle-rank SoftImpute

Oracle rank.max = 6; effective rank = 6; lambda = 1; RMSE = 0.2424



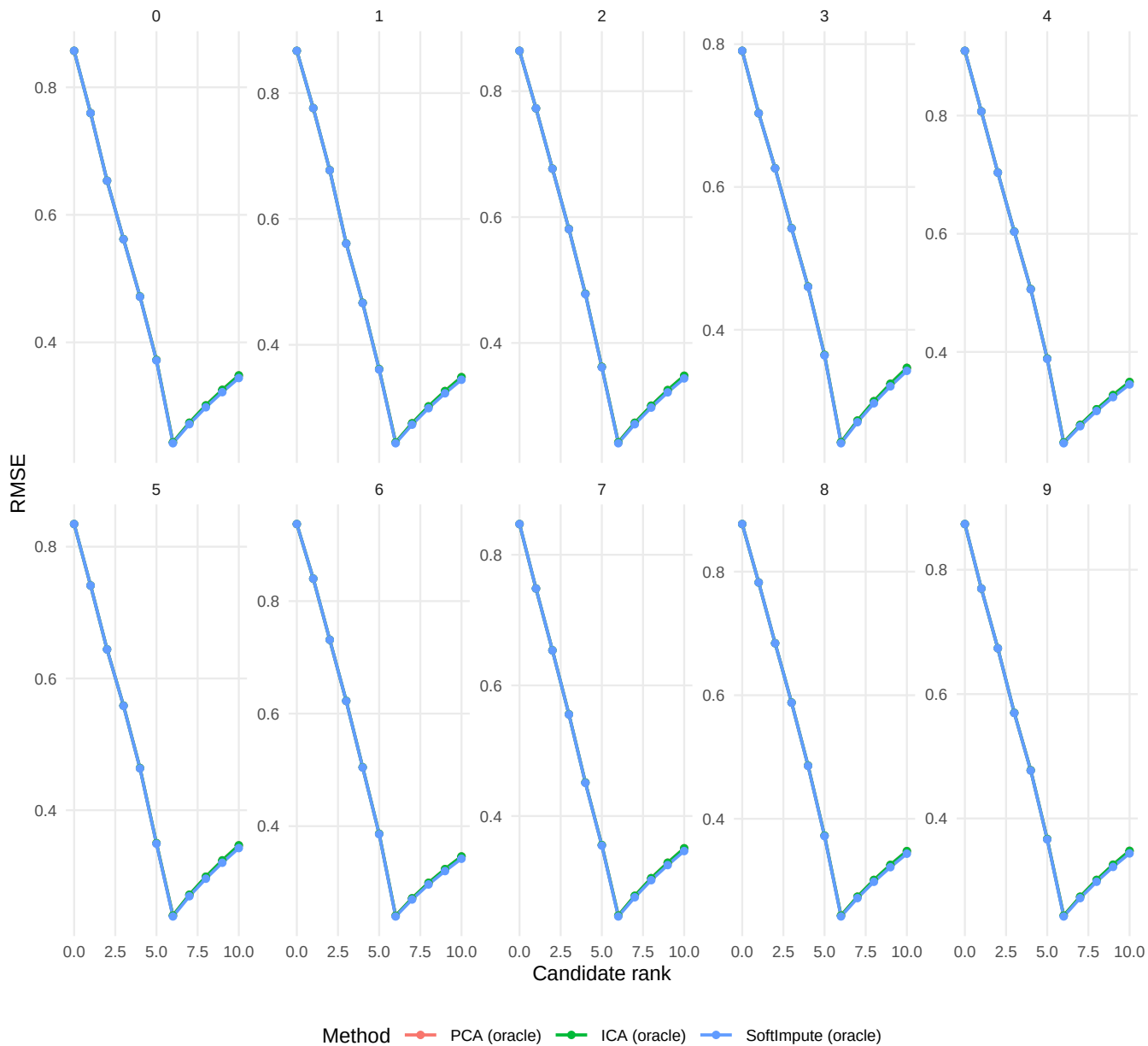
Denoising RMSE by simulation, independant binary simulation

RMSE no denoising = 1.251



Oracle reconstruction RMSE by candidate rank

SoftImpute uses fixed lambda = 1



Method PCA (oracle) ICA (oracle) SoftImpute (oracle)